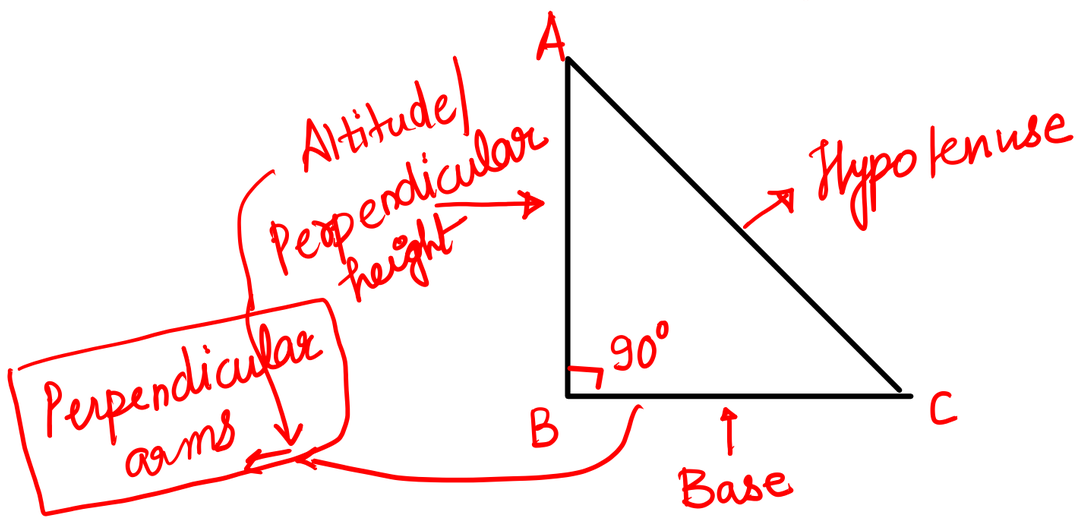
INTRODUCTION TO TRIGONOMETRY 8

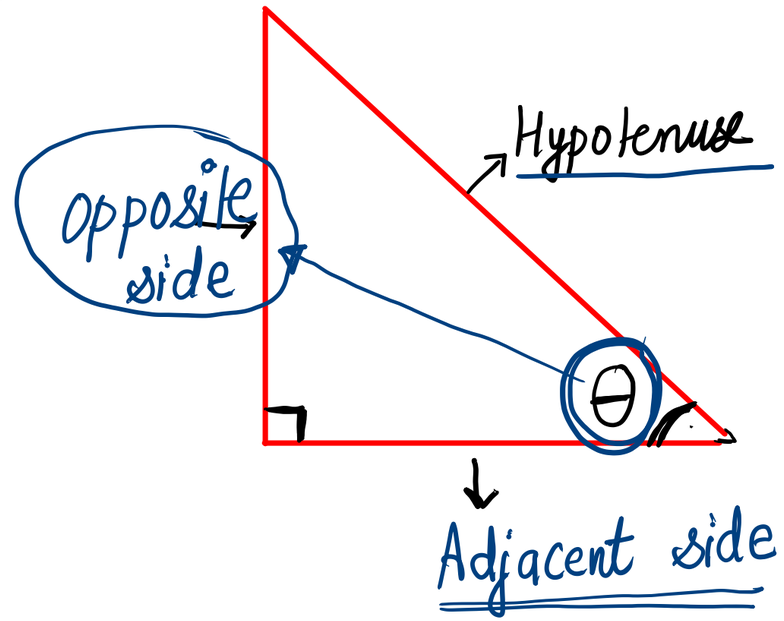
TRIGONOMETRY

- TRI → Three
- GONO → Sides
- METRY → Measure

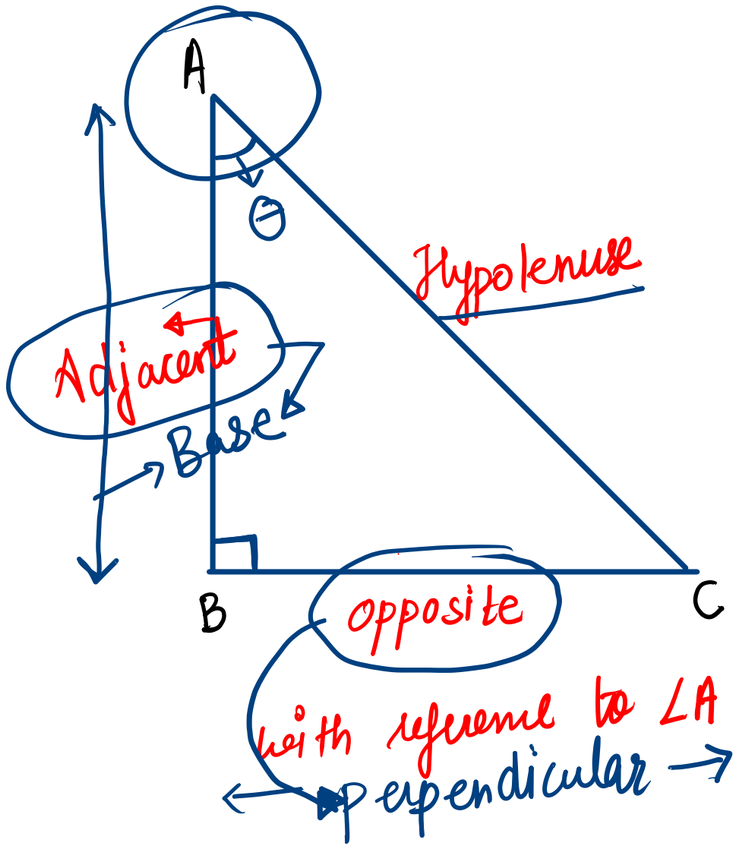
Triangle

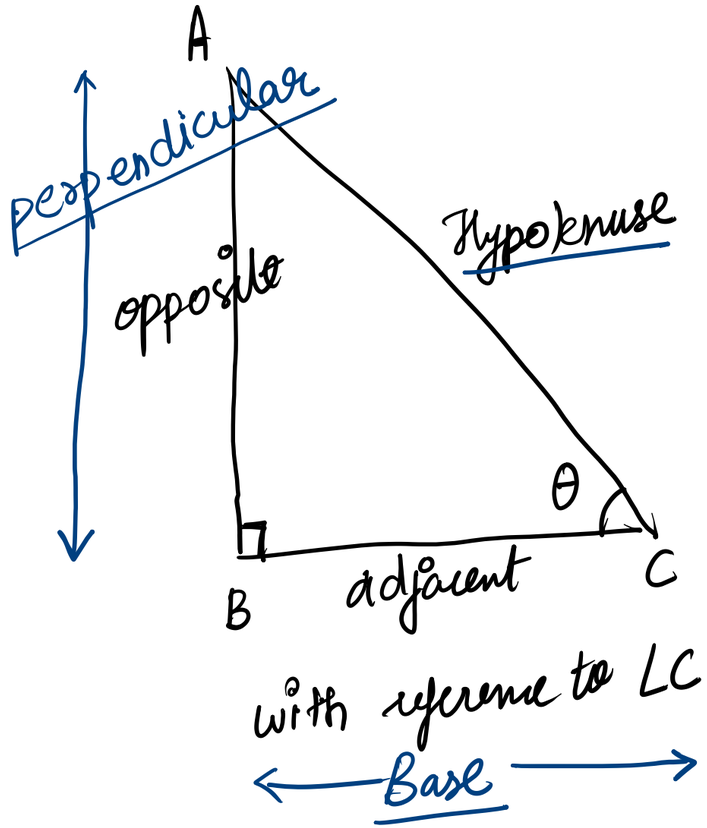
→ Right angle Triangle





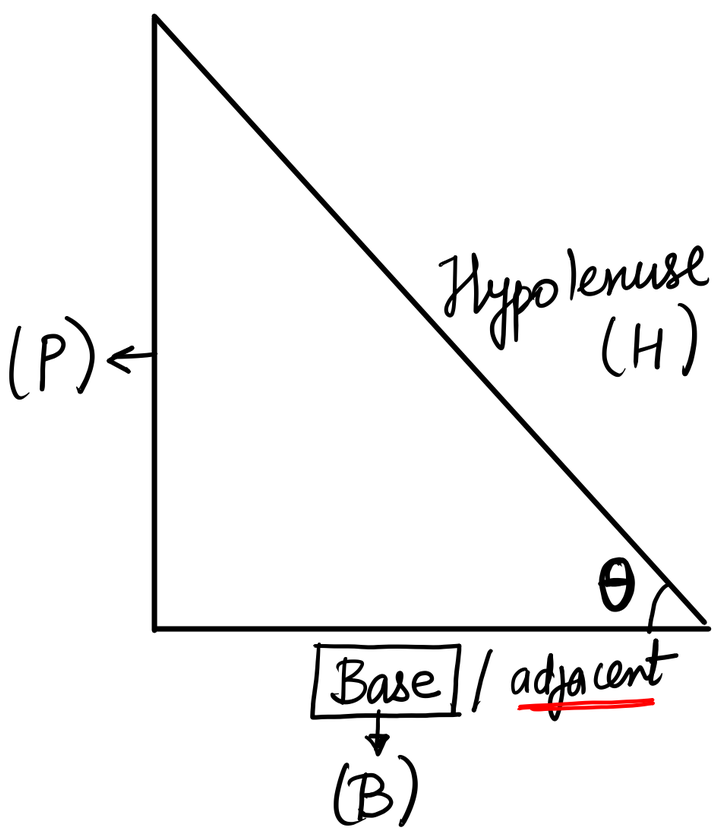
θ → theta

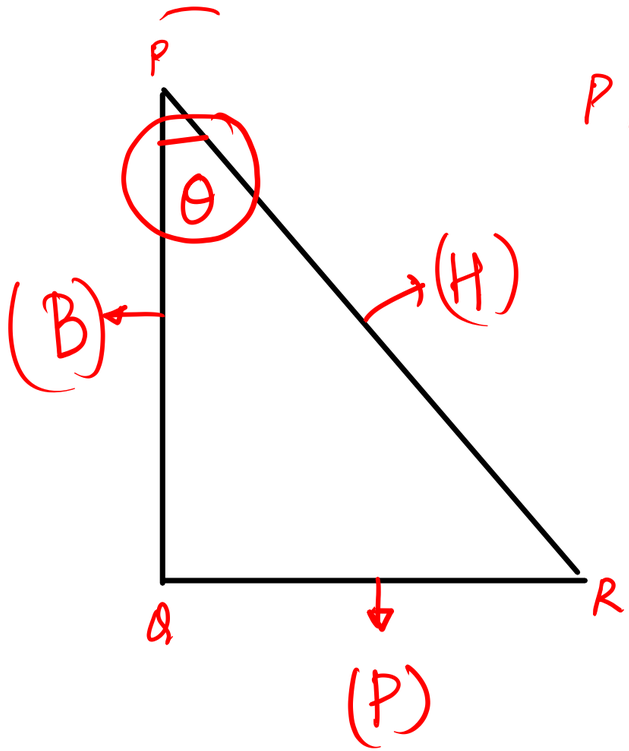




opposite → perpendicular

adjacent → Base

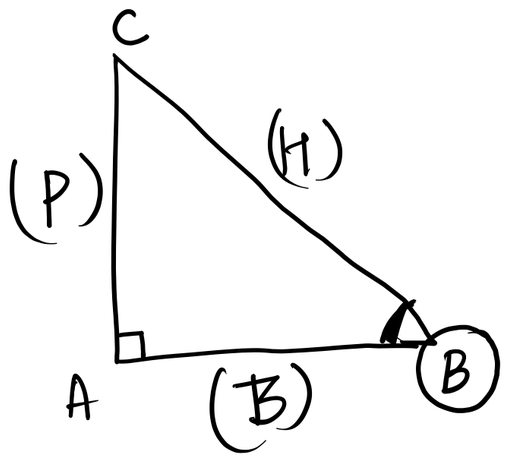




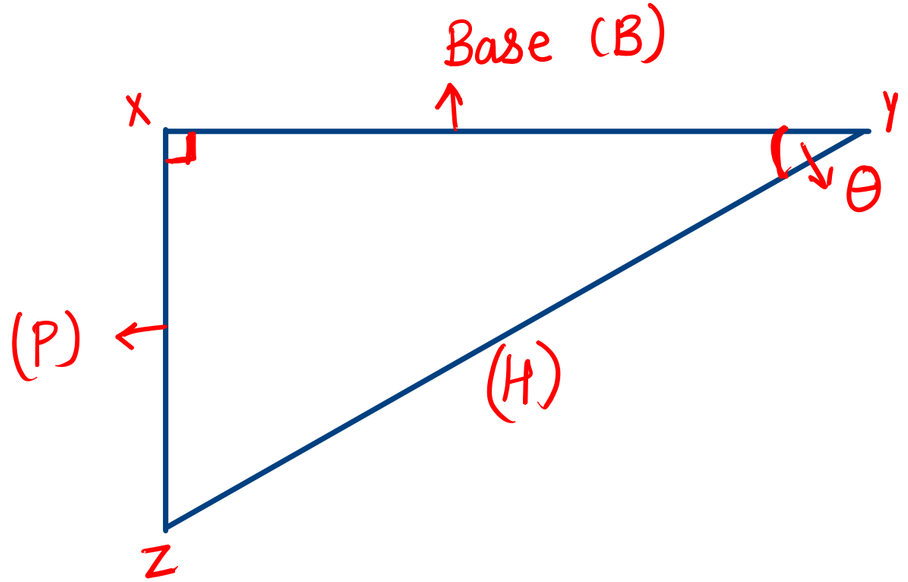
P, B, H

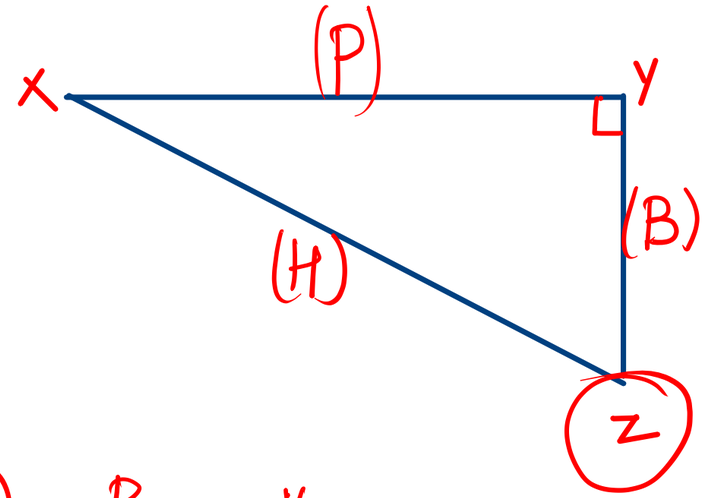
Opposite → P

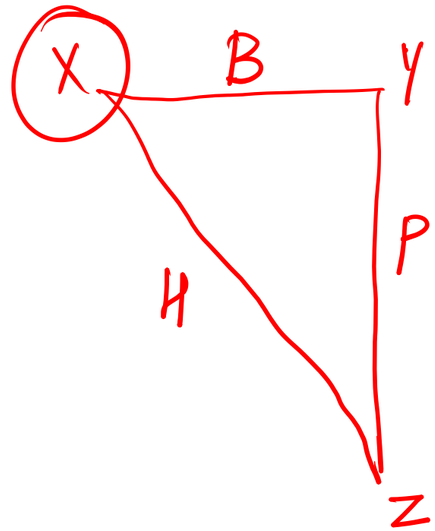
Adjacent → B

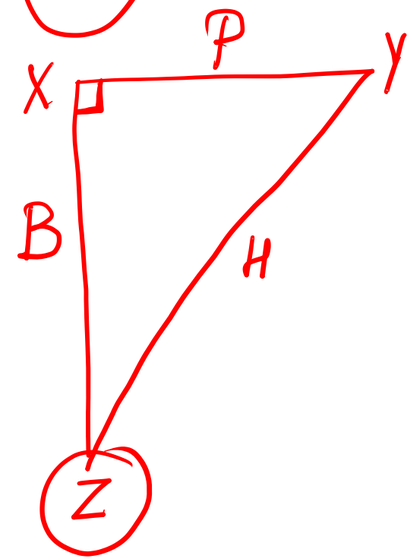


P, B, H

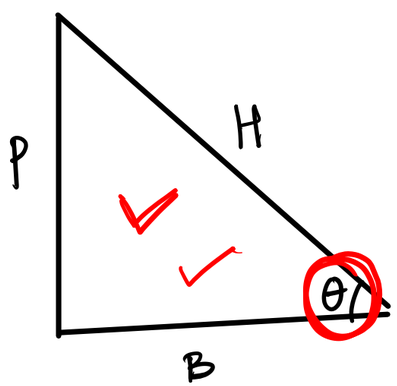








TRIGONOMETRIC RATIOS



Sine $\theta = \frac{P}{H}$

cosine $\theta = \frac{B}{H}$

tangent $\theta = \frac{P}{B}$

cotangent $\theta = \frac{B}{P}$

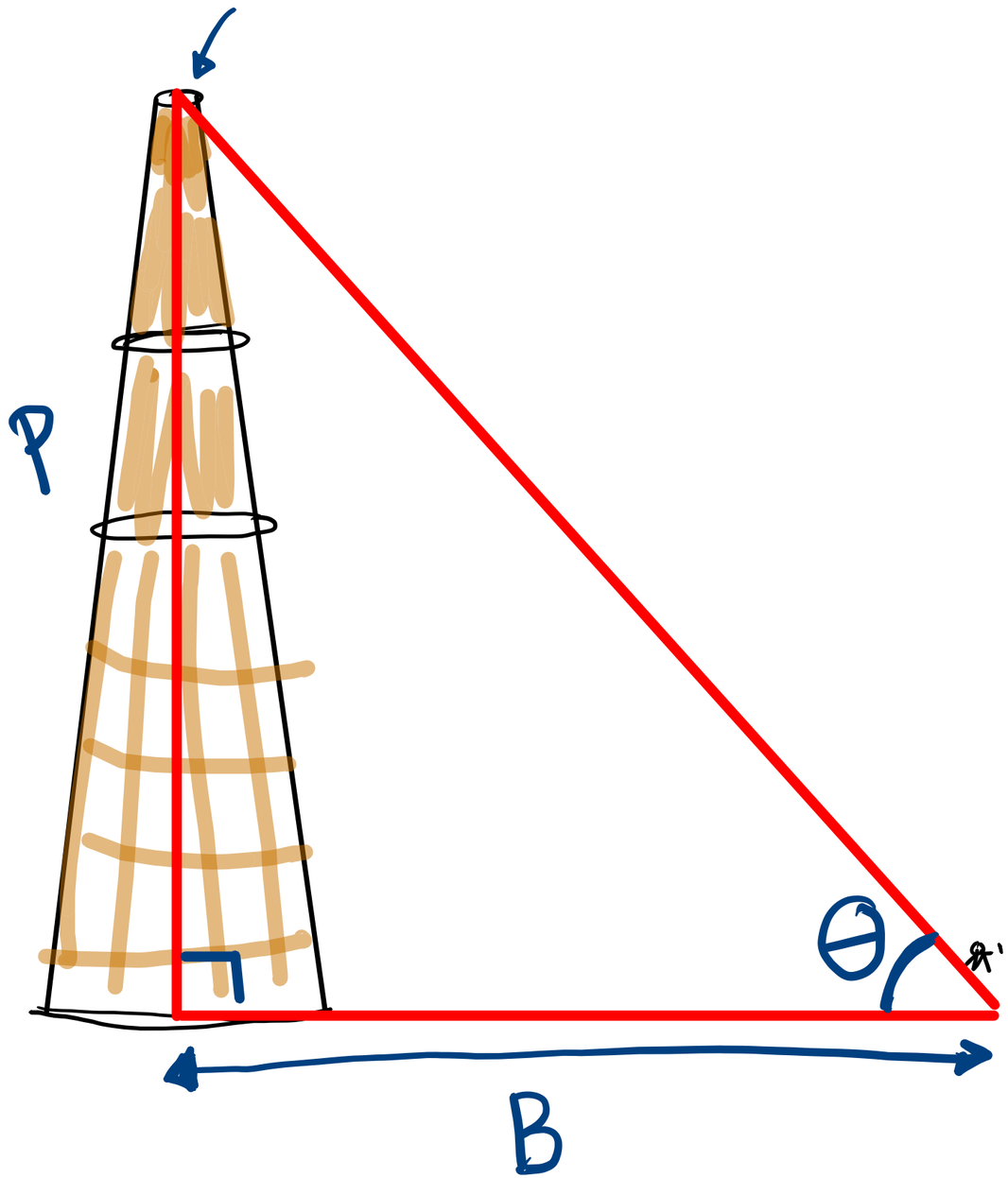
cosecant $\theta = \frac{H}{P}$

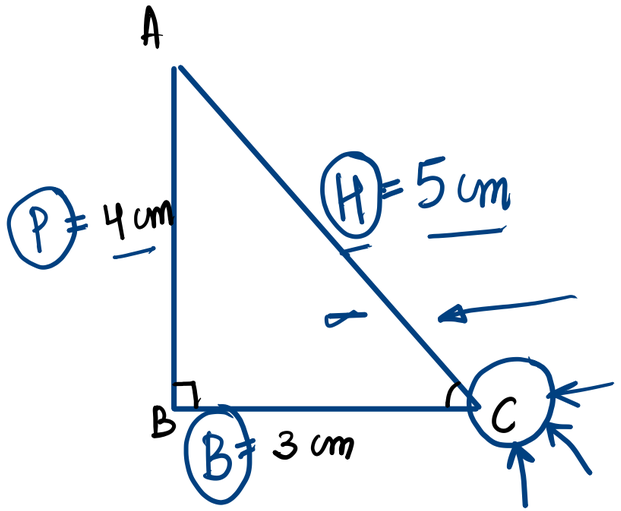
Secant $\theta = \frac{H}{B}$

$\sin\theta = \frac{P}{H}$ $\quad$ $\operatorname{cosec}\theta = \frac{H}{P}$

$\cos\theta = \frac{B}{H}$ $\quad$ $\sec\theta = \frac{H}{B}$

$\tan\theta = \frac{P}{B}$ $\quad$ $\cot\theta = \frac{B}{P}$


P
θ
B



Find $\sin C = \frac{P}{H}$

$= \frac{4}{5}$

(ii) $\cos C = \frac{B}{H} = \frac{3}{5}$

$\tan C = \frac{4}{3}$

$\cot C = \frac{B}{P}$

$= \frac{3}{4}$

$\cot A = \frac{4}{3}$

$\sec A = \frac{H}{B} = \frac{5}{4}$

$\tan A = \frac{P}{B}$

$= \frac{3}{4}$

$\cos A = \frac{B}{H} = \frac{4}{5}$

Pythagoras theorem

$P^2 + B^2 = H^2$

$4^2 + 3^2 = H^2$

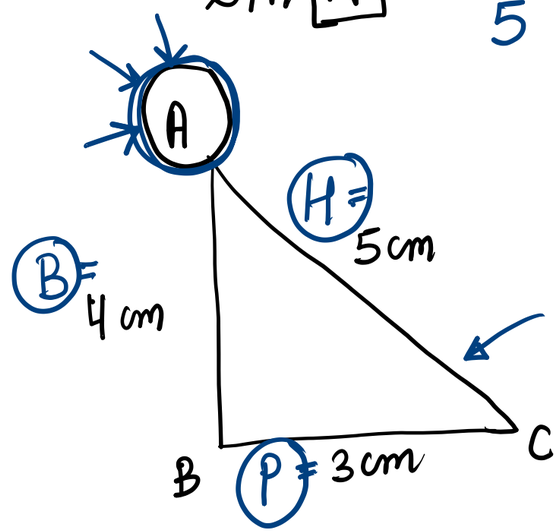
$16 + 9 = H^2$

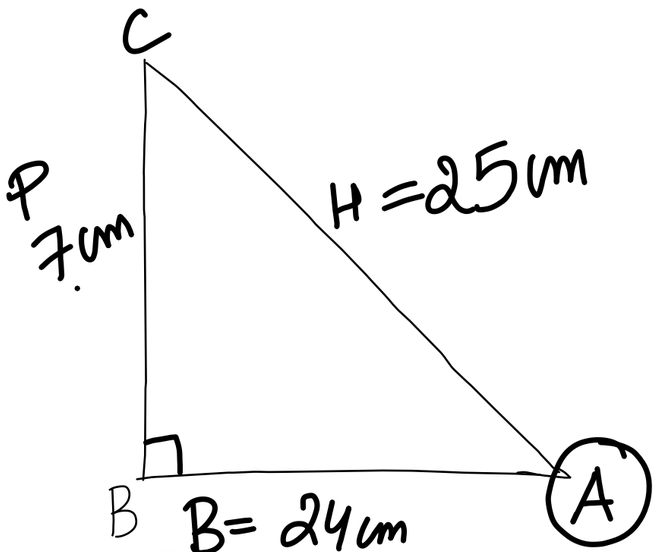
$25 = H^2$

$5^2 = H^2$

$H = 5$ cm

$\sin A = \frac{3}{5}$





(i) $\sin A$, $\cos A$

Apply the pythagoras theorem

$$H^2 = P^2 + B^2$$

$$H^2 = 7^2 + 24^2$$

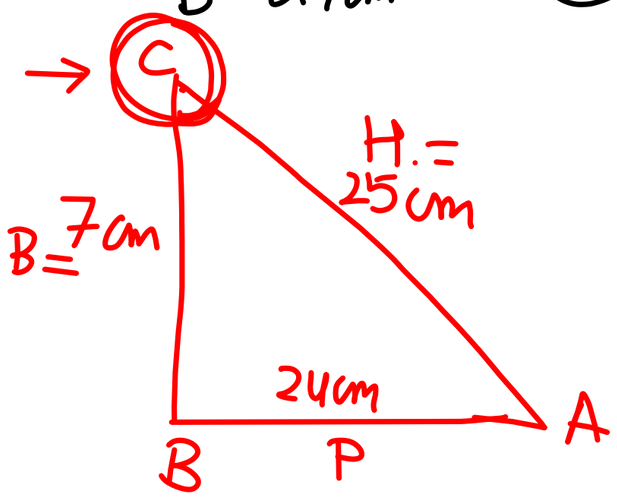
$$H^2 = 49 + 576$$

$$H^2 = 625$$

$$H = \sqrt{625} = 25$$

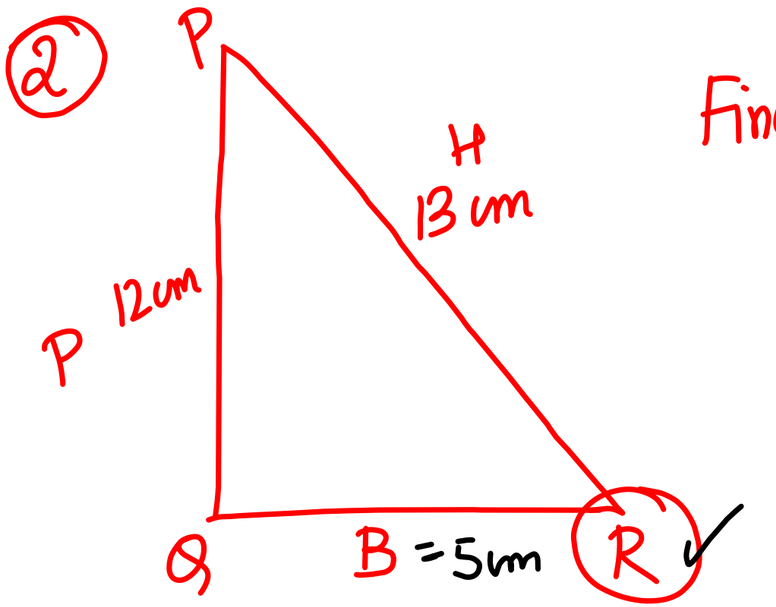
$$\sin A = \frac{P}{H} = \frac{7}{25}$$

$$\cos A = \frac{24}{25}$$



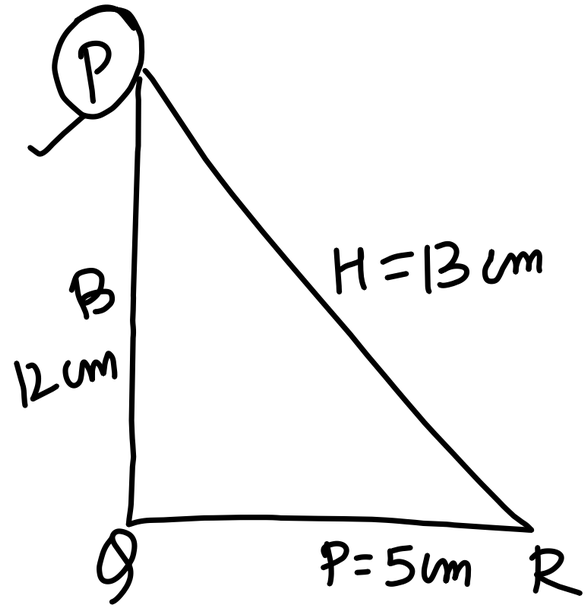
(ii) $\sin C$, $\cos C$

$$\sin C = \frac{24}{25} \qquad \cos C = \frac{7}{25}$$

② 

Find $\tan P - \cot R$

$\frac{5}{12} - \frac{5}{12} = 0$



$H^2 = P^2 + B^2$

$13^2 = 12^2 + B^2$

$169 = 144 + B^2$

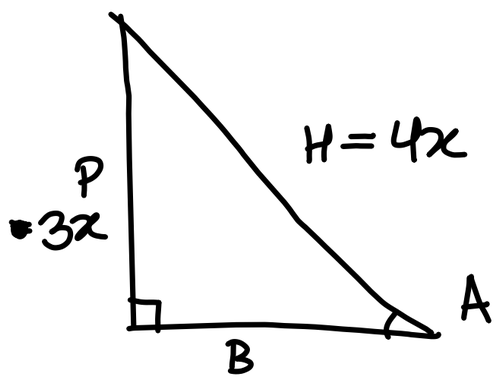
$B^2 = 25$

$B = 5$

$\tan P = \frac{5}{12}$

$\cot R = \frac{5}{12}$

(3) If $\sin A = \frac{3}{4}$, calculate $\cos A$ & $\tan A$.

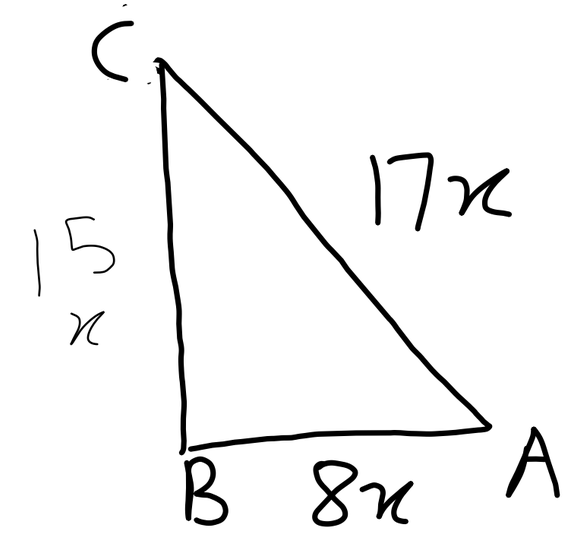


$H^2 = P^2 + B^2$

$(4x)^2 =$

④ Given $15 \cot A = 8$, find $\sin A$ & $\sec A$.

$$\cot A = \frac{8}{15} = \frac{B}{P}$$



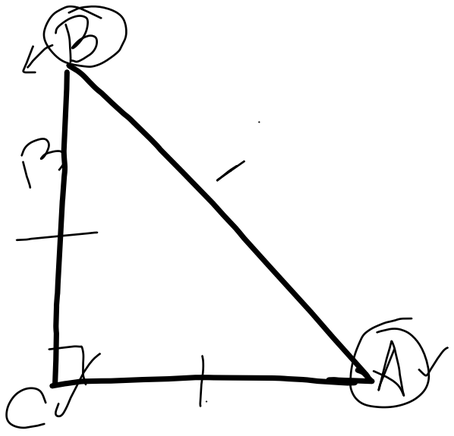
$$H^2 = P^2 + B^2$$
$$H^2 = (15x)^2 + (8x)^2$$
$$H^2 = 225x^2 + 64x^2$$
$$H^2 = 289x^2$$
$$H = \sqrt{289x^2}$$
$$H = 17x$$

$$\sin A = \frac{P}{H} = \frac{15x}{17x} = \frac{15}{17}$$

$$\sec A = \frac{H}{B} = \frac{17x}{8x} = \frac{17}{8}$$

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⑥ $\angle A, \angle B$ are acute angles such that $\cos A = \cos B$, then show $\angle A = \angle B$.



$\cos A = \frac{AC}{AB}$

$\cos B = \frac{BC}{AB}$

Given $\cos A = \cos B$

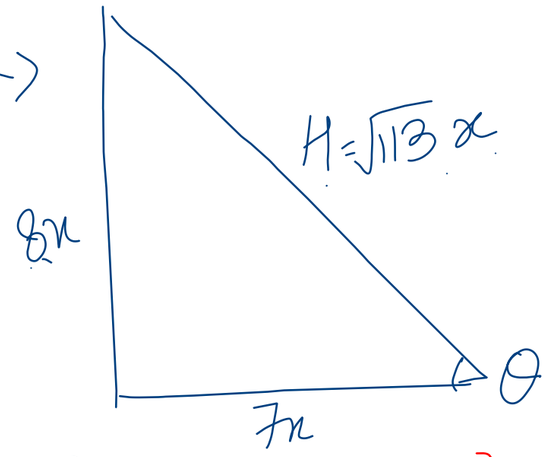
$\frac{AC}{AB} = \frac{BC}{AB}$

$AC = BC$

So $\triangle ABC$ is an isosceles $\triangle$.

Also, angles opposite to the equal sides are equal $\boxed{\angle A = \angle B}$

⑦ $\cot\theta = \frac{7}{8}$ → B, P



$P^2 + B^2 = H^2$

$(8x)^2 + (7x)^2 = H^2$

$64x^2 + 49x^2 = H^2$

$113x^2 = H^2$

$\sqrt{113}x = H$

$\cot^2\theta = (\cot\theta)^2$

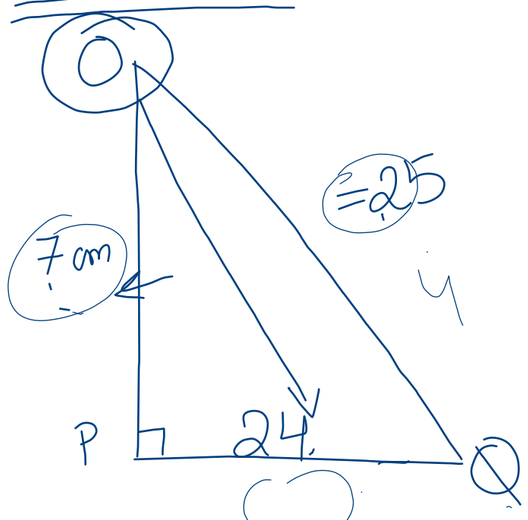
$a^2 - b^2 = (a+b)(a-b)$

i) $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$

$\sin\theta = \frac{8}{\sqrt{113}}$ $\quad \cos\theta = \frac{7}{\sqrt{113}}$

$$\frac{\left(1+\frac{8}{\sqrt{113}}\right)\left(1-\frac{8}{\sqrt{113}}\right)}{\left(1+\frac{7}{\sqrt{113}}\right)\left(1-\frac{7}{\sqrt{113}}\right)} = \frac{1^2-\left(\frac{8}{\sqrt{113}}\right)^2}{1^2-\left(\frac{7}{\sqrt{113}}\right)^2} = \frac{1-\frac{64}{113}}{1-\frac{49}{113}} = \frac{\frac{49}{113}}{\frac{64}{113}} = \frac{49}{64}$$

Example 5



$OP = 7\ cm$

$OQ - PQ = 1\ cm$

find $\sin Q$ & $\cos Q$

$\sin Q = \frac{7}{25}$ $\sin Q = \frac{24}{25}$

$\cos Q = \frac{24}{25}$ $\tan Q = \frac{24}{7}$

Apply Pythagoras theorem

$OP^2 + PQ^2 = OQ^2$

$(7)^2 + (OQ - 1)^2 = OQ^2$

$49 + (OQ)^2 + (1)^2 - 2OQ = OQ^2$

$50 = 2OQ$

$OQ = 25$

625

$\sin Q = \frac{P}{H}$ $\frac{B}{H}$ $\frac{P}{B}$

10. In Δ PQR, right-angled at Q, PR + QR = 25 cm and PQ = 5 cm. Determine the values of sin P, cos P and tan P.

HW

11. State whether the following are true or false. Justify your answer.

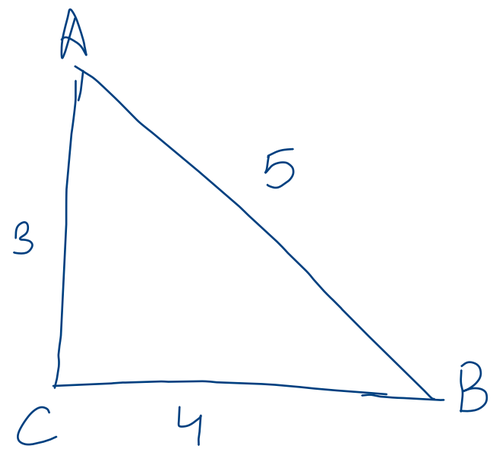
(i) The value of tan A is always less than 1. False

(ii) $\sec A = \frac{12}{5}$ for some value of angle A. True $\frac{H}{B}$

(iii) cos A is the abbreviation used for the cosecant of angle A. False

(iv) cot A is the product of cot and A. False

(v) $\sin\theta = \frac{4}{3}$ for some angle θ. False



$\tan A = \frac{4}{3} =$

$\tan B = \frac{3}{4}$

not to scale

$$\operatorname{cosec}\theta = \frac{1}{\sin\theta} \quad \Rightarrow \quad \sin\theta = \frac{1}{\operatorname{cosec}\theta}$$

$$\sec\theta = \frac{1}{\cos\theta} \quad \Rightarrow \quad \cos\theta = \frac{1}{\sec\theta}$$

$$\cot\theta = \frac{1}{\tan\theta} \quad \Rightarrow \quad \tan\theta = \frac{1}{\cot\theta}$$

* $$\boxed{\tan\theta = \frac{\sin\theta}{\cos\theta}}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

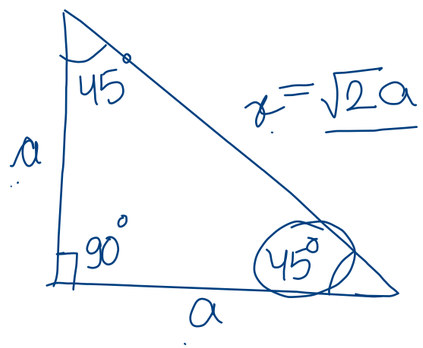
8.3 Trigonometric Ratios of Some Specific Angles

13/04/26

From geometry, you are already familiar with the construction of angles of 30°, 45°, 60° and 90°. In this section, we will find the values of the trigonometric ratios for these angles and, of course, for 0°.

$\angle A$	0°	30°	45°	60°	90°
sin A	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos A	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan A	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
cosec A	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec A	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
cot A	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

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$$\sin 45^\circ = \frac{a}{\sqrt{2}a} = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = \frac{a}{a} = 1$$

$$a^2 + a^2 = x^2$$

$$2a^2 = x^2$$

$$\sqrt{2a^2} = \sqrt{x^2}$$

$$\sqrt{2}a = x$$

$\frac{\sqrt{0}}{2}$ $\frac{\sqrt{1}}{2}$ $\frac{\sqrt{2}}{2}$ $\frac{\sqrt{3}}{2}$ $\frac{\sqrt{4}}{2}$

$= 0$ $\frac{1}{2}$ $\frac{1}{\sqrt{2}}$ $\frac{\sqrt{3}}{2}$ 1

$\frac{\sqrt{2}}{\sqrt{2}\times\sqrt{2}}$

	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1 →
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0 ←
$\tan\theta$ ← $\frac{\sin\theta}{\cos\theta}$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	N.D →
$\cot\theta$	N.D	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0 ←
$\operatorname{cosec}\theta$ ← $\frac{1}{\sin\theta}$	N.D	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1 →
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	N.D ←

$\frac{x}{0} = \text{N.D.}$

EXERCISE 8.2

1. Evaluate the following :

(i) $\sin 60° \cos 30° + \sin 30° \cos 60°$

(ii) $2 \tan^2 45° + \cos^2 30° - \sin^2 60°$

(iii) $\dfrac{\cos 45°}{\sec 30° + \operatorname{cosec} 30°}$

(iv) $\dfrac{\sin 30° + \tan 45° - \operatorname{cosec} 60°}{\sec 30° + \cos 60° + \cot 45°}$

(v) $\dfrac{5 \cos^2 60° + 4 \sec^2 30° - \tan^2 45°}{\sin^2 30° + \cos^2 30°}$

(1) i) $\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2}$

$\Rightarrow \frac{3}{4} + \frac{1}{4}$

$= \frac{4}{4} = 1$

ii) $2(\tan 45)^2 + (\cos 30)^2 - (\sin 60)^2$

$= 2(1)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2$

$= 2 + \frac{3}{4} - \frac{3}{4}$

$= 2$

EXERCISE 8.2

1. Evaluate the following :

(i) $\sin 60° \cos 30° + \sin 30° \cos 60°$

(ii) $2 \tan^2 45° + \cos^2 30° - \sin^2 60°$

(iii) $\dfrac{\cos 45°}{\sec 30° + \operatorname{cosec} 30°}$

(iv) $\dfrac{\sin 30° + \tan 45° - \operatorname{cosec} 60°}{\sec 30° + \cos 60° + \cot 45°}$

(v) $\dfrac{5 \cos^2 60° + 4 \sec^2 30° - \tan^2 45°}{\sin^2 30° + \cos^2 30°}$

(iv) $\dfrac{\frac{1}{2} + 1 - \frac{2}{\sqrt{3}}}{\frac{2}{\sqrt{3}} + \frac{1}{2} + \frac{1}{1}} = \dfrac{\frac{\sqrt{3} + 2\sqrt{3} - 4}{2\sqrt{3}}}{\frac{4 + \sqrt{3} + 2\sqrt{3}}{2\sqrt{3}}} = \dfrac{3\sqrt{3} - 4}{4 + 3\sqrt{3}} \times \dfrac{4 - 3\sqrt{3}}{4 - 3\sqrt{3}}$

$= \dfrac{12\sqrt{3} - 27 - 16 + 12\sqrt{3}}{4^2 - (3\sqrt{3})^2} = \dfrac{-43 + 24\sqrt{3}}{16 - 27} = \dfrac{-43 + 24\sqrt{3}}{-11} = \dfrac{43 - 24\sqrt{3}}{11}$

2. Choose the correct option and justify your choice :

(i) $\dfrac{2 \tan 30°}{1 + \tan^2 30°} =$

(A) $\sin 60°$ (B) $\cos 60°$ (C) $\tan 60°$ (D) $\sin 30°$

(ii) $\dfrac{1 - \tan^2 45°}{1 + \tan^2 45°} =$

(A) $\tan 90°$ (B) 1 (C) $\sin 45°$ (D) 0

(iii) $\sin 2A = 2 \sin A$ is true when A =

(A) $0°$ (B) $30°$ (C) $45°$ (D) $60°$

(iv) $\dfrac{2 \tan 30°}{1 - \tan^2 30°} =$

(A) $\cos 60°$ (B) $\sin 60°$ (C) $\tan 60°$ (D) $\sin 30°$

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21/04/26

3. If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \frac{1}{\sqrt{3}}$; $0° < A+B \leq 90°$; $A > B$, find A and B.

Sol^n

$\tan(A+B) = \sqrt{3}$

We know that $\tan 60° = \sqrt{3}$

So, $A+B = 60$ — (i)

$\tan(A-B) = \frac{1}{\sqrt{3}}$

$\tan 30° = \frac{1}{\sqrt{3}}$

$A-B = 30°$ — (ii)

Add (i) & (ii)

$A+B+A-B = 60+30$

$2A = 90$

$A = 45°$

$45+B = 60$

$B = 15°$

4. State whether the following are true or false. Justify your answer.

(i) $\sin(A+B) = \sin A + \sin B$. F

→ (ii) The value of $\sin\theta$ increases as θ increases. T

(iii) The value of $\cos\theta$ increases as θ increases. F

(iv) $\sin\theta = \cos\theta$ for all values of θ. F

(v) $\cot A$ is not defined for $A = 0°$. T

(i) $\sin(30+60) \stackrel{?}{=} \sin 30 + \sin 60$

$1 \stackrel{?}{=} \frac{1}{2} + \frac{\sqrt{3}}{2}$

8.4 Trigonometric Identities

You may recall that an equation is called an identity when it is true for all values of the variables involved. Similarly, an equation involving trigonometric ratios of an angle is called a **trigonometric identity,** if it is true for all values of the angle(s) involved.

In this section, we will prove one trigonometric identity, and use it further to prove other useful trigonometric identities.

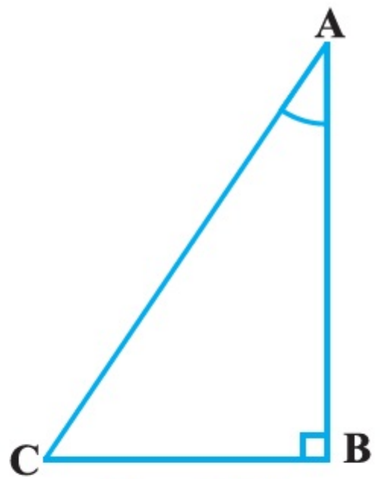


Fig. 8.21

In ΔABC, right-angled at B (see Fig. 8.21), we have:

$$AB^2 + BC^2 = AC^2 \quad (1)$$

Dividing each term of (1) by AC^2, we get

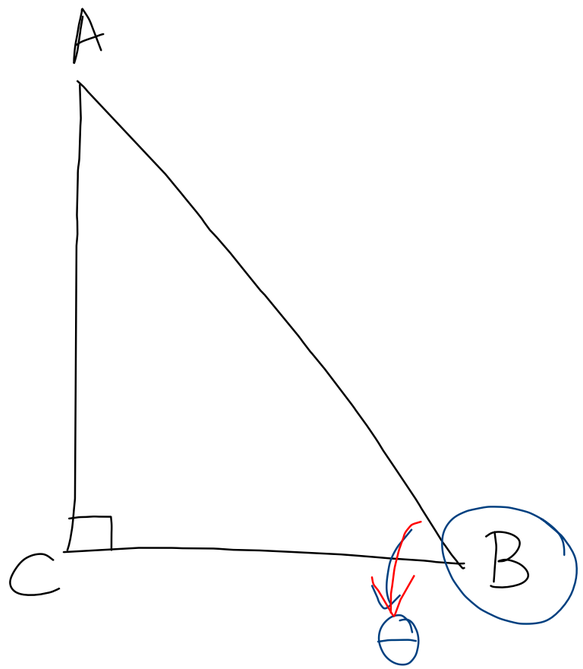
$$\frac{AB^2}{AC^2} + \frac{BC^2}{AC^2} = \frac{AC^2}{AC^2}$$

i.e., $$\left(\frac{AB}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = \left(\frac{AC}{AC}\right)^2$$

i.e., $$(\cos A)^2 + (\sin A)^2 = 1$$

i.e., $$\mathbf{\cos^2 A + \sin^2 A = 1} \quad (2)$$

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$$AC^2 + BC^2 = AB^2$$

Divide both sides by AB^2

$$\frac{AC^2}{AB^2} + \frac{BC^2}{AB^2} = \frac{AB^2}{AB^2}$$

$$\Rightarrow \left(\frac{AC}{AB}\right)^2 + \left(\frac{BC}{AB}\right)^2 = 1$$

$$\Rightarrow \boxed{\sin^2 B + \cos^2 B = 1}$$

$$\boxed{\sin^2\theta + \cos^2\theta = 1}$$

Divide both sides by BC^2

$$\left(\frac{AC}{BC}\right)^2 + \left(\frac{BC}{BC}\right)^2 = \left(\frac{AB}{BC}\right)^2$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$\Rightarrow \boxed{1 + \tan^2\theta = \sec^2\theta}$$

Divide both sides by AC^2

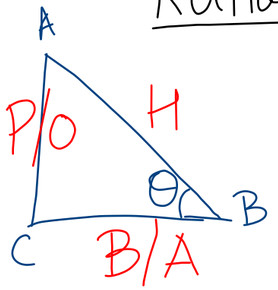
$$\left(\frac{AC}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = \left(\frac{AB}{AC}\right)^2$$

$$\Rightarrow \boxed{1 + \cot^2\theta = \operatorname{cosec}^2\theta}$$

Example 9 : Express the ratios cos A, tan A and sec A in terms of sin A.

Example 10 : Prove that sec A (1 – sin A)(sec A + tan A) = 1.

Ratios



$\sin\theta = \frac{P}{H}$; $\operatorname{cosec}\theta = \frac{H}{P}$

$\cos\theta = \frac{B}{H}$; $\sec\theta = \frac{H}{B}$

$\tan\theta = \frac{P}{B}$; $\cot\theta = \frac{B}{P}$

Specific Angles

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	N.D
cosec	N.D	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	N.D
cot	N.D	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

All Important Ratio & Formulaes of Trigonometry Chapter

$\frac{1}{\sin\theta} = \operatorname{cosec}\theta$; $\frac{1}{\operatorname{cosec}\theta} = \sin\theta$

$\frac{1}{\cos\theta} = \sec\theta$; $\frac{1}{\sec\theta} = \cos\theta$

$\tan\theta = \frac{\sin\theta}{\cos\theta}$; $\cot\theta = \frac{\cos\theta}{\sin\theta}$

$\frac{1}{\tan\theta} = \cot\theta$

$\frac{1}{\cot\theta} = \tan\theta$

Identities

① $\sin^2\theta + \cos^2\theta = 1$.

- $1 - \sin^2\theta = \cos^2\theta$
- $1 - \cos^2\theta = \sin^2\theta$

② $1 + \tan^2\theta = \sec^2\theta$

- $\sec^2\theta - \tan^2\theta = 1$
- $\sec^2\theta - 1 = \tan^2\theta$

③ $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.

- $\operatorname{cosec}^2\theta - 1 = \cot^2\theta$
- $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$1+\cot^2\theta = \operatorname{cosec}^2\theta$

$1+\cot^2\theta = \frac{1}{\sin^2\theta}$

$\sin^2\theta = \frac{1}{1+\cot^2\theta}$

$\sin\theta = \frac{1}{\sqrt{1+\cot^2\theta}}$

(i) $\sin\theta =$ ☐

$1+\tan^2\theta = \sec^2\theta$

$\sqrt{1+\tan^2\theta} = \sec\theta$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$

$\sin\theta = \frac{\cos\theta}{\cot\theta} = \frac{1}{\cot\theta}\cdot\cos\theta$

$= \frac{1}{\cot\theta}\cdot\frac{1}{\sec\theta} = \frac{1}{\cot\theta\cdot\sec\theta} = \frac{1}{\cot\theta\cdot\sqrt{1+\tan^2\theta}}$

$\frac{1}{\cos^2 A} \div \frac{1}{\sin^2 A} = \frac{1}{\cos^2 A}\times\frac{\sin^2 A}{1} =$

EXERCISE 8.3

1. Express the trigonometric ratios sin A, sec A and tan A in terms of cot A.
2. Write all the other trigonometric ratios of $\angle$ A in terms of sec A.
3. Choose the correct option. Justify your choice.

(i) $9\sec^2 A - 9\tan^2 A =$ $9(\sec^2 A - \tan^2 A) = 9(1) = 9$

(A) 1 (B) 9 (C) 8 (D) 0

(ii) $(1+\tan\theta+\sec\theta)(1+\cot\theta-\operatorname{cosec}\theta) =$

(A) 0 (B) 1 (C) 2 (D) –1

(iii) $(\sec A+\tan A)(1-\sin A) =$

(A) sec A (B) sin A (C) cosec A (D) cos A

(iv) $\frac{1+\tan^2 A}{1+\cot^2 A} = \frac{\sec^2 A}{\operatorname{cosec}^2 A} = \frac{\frac{1}{\cos^2 A}}{\frac{1}{\sin^2 A}} = \frac{\sin^2 A}{\cos^2 A} = \left(\frac{\sin A}{\cos A}\right)^2 = \tan^2 A$

(A) $\sec^2 A$ (B) –1 (C) $\cot^2 A$ (D) $\tan^2 A$

(ii) $\left(1+\frac{\sin\theta}{\cos\theta}+\frac{1}{\cos\theta}\right)\left(\frac{1}{1}+\frac{\cos\theta}{\sin\theta}-\frac{1}{\sin\theta}\right)$

$\Rightarrow \left(\frac{\cos\theta+\sin\theta+1}{\cos\theta}\right)\left(\frac{\sin\theta+\cos\theta-1}{\sin\theta}\right)$

$\Rightarrow \frac{(\sin\theta+\cos\theta+1)(\sin\theta+\cos\theta-1)}{\sin\theta\cdot\cos\theta}$

$\Rightarrow \frac{(\sin\theta+\cos\theta)^2-1}{\sin\theta\cdot\cos\theta} = \frac{\sin^2\theta+\cos^2\theta+2\sin\theta\cos\theta-1}{\sin\theta\cdot\cos\theta}$

$= \frac{2\sin\theta\cdot\cos\theta}{\sin\theta\cdot\cos\theta}$

$= 2$

$\frac{1}{6}+\frac{1}{6} =$

$\frac{1}{1}+\frac{1}{6}+\frac{2}{6} =$

(iii) $\left(\frac{1}{\cos A}+\frac{\sin A}{\cos A}\right)(1-\sin A)$

$\Rightarrow \frac{(1+\sin A)(1-\sin A)}{\cos A}$

$= \frac{1-\sin^2 A}{\cos A}$

$= \frac{\cos^2 A}{\cos A} = \cos A$ (D)

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4. Prove the following identities, where the angles involved are acute angles for which the expressions are defined.

(i) $(\text{cosec}\,\theta - \cot\theta)^2 = \dfrac{1-\cos\theta}{1+\cos\theta}$

(ii) $\dfrac{\cos A}{1+\sin A} + \dfrac{1+\sin A}{\cos A} = 2\sec A$

☆ (iii) $\dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\ \text{cosec}\,\theta$

[**Hint :** Write the expression in terms of sin θ and cos θ]

(iii) $\dfrac{\frac{\sin\theta}{\cos\theta}}{1-\frac{\cos\theta}{\sin\theta}} + \dfrac{\frac{\cos\theta}{\sin\theta}}{1-\frac{\sin\theta}{\cos\theta}}$

LHS =

$\Rightarrow \dfrac{\frac{\sin\theta}{\cos\theta}}{\frac{\sin\theta-\cos\theta}{\sin\theta}} + \dfrac{\frac{\cos\theta}{\sin\theta}}{\frac{\cos\theta-\sin\theta}{\cos\theta}}$

$\Rightarrow \dfrac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} + \dfrac{\cos^2\theta}{\sin\theta(\cos\theta-\sin\theta)}$

$\Rightarrow \dfrac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} - \dfrac{\cos^2\theta}{\sin\theta(\sin\theta-\cos\theta)}$

$\dfrac{1}{\sin\theta-\cos\theta}\left[\dfrac{\sin^2\theta}{\cos\theta} - \dfrac{\cos^2\theta}{\sin\theta}\right]$

$\dfrac{1}{\sin\theta-\cos\theta}\left[\dfrac{\sin^3\theta-\cos^3\theta}{\sin\theta\cdot\cos\theta}\right]$

$\dfrac{1}{(\sin\theta-\cos\theta)}\left[\dfrac{(\sin\theta-\cos\theta)(\sin^2\theta+\cos^2\theta+\sin\theta\cdot\cos\theta)}{\sin\theta\cdot\cos\theta}\right]$

$a^3-b^3 = (a-b)(a^2+b^2+ab)$

$\dfrac{a+b}{c} = \dfrac{a}{c}+\dfrac{b}{c}$

$\dfrac{1}{b-a} = \dfrac{1}{-(a-b)}$

$\dfrac{1}{\cos\theta-\sin\theta} = \dfrac{1}{-(\sin\theta-\cos\theta)}$

$\dfrac{1+\sin\theta\cos\theta}{\sin\theta\cdot\cos\theta} = \dfrac{1}{\sin\theta\cdot\cos\theta} + \dfrac{\sin\theta\cdot\cos\theta}{\sin\theta\cdot\cos\theta}$

$= \dfrac{1}{\sin\theta}\cdot\dfrac{1}{\cos\theta} + 1$

$= \text{cosec}\theta\cdot\sec\theta + 1$

$= 1 + \sec\theta\cdot\text{cosec}\theta = \text{RHS}$

(iv) $\dfrac{1+\sec A}{\sec A} = \dfrac{\sin^2 A}{1-\cos A}$ [**Hint :** Simplify LHS and RHS separately]

(v) $\dfrac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A$, using the identity $\operatorname{cosec}^2 A = 1 + \cot^2 A$.

(vi) $\sqrt{\dfrac{1+\sin A}{1-\sin A}} = \sec A + \tan A$ (vii) $\dfrac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$

(viii) $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$

(vi) $\sqrt{\dfrac{1+\sin A}{1-\sin A} \times \dfrac{1+\sin A}{1+\sin A}}$

$= \sqrt{\dfrac{(1+\sin A)^2}{1-\sin^2 A}}$

$= \sqrt{\dfrac{(1+\sin A)^2}{\cos^2 A}} = \dfrac{1+\sin A}{\cos A}$

$= \dfrac{1}{\cos A} + \dfrac{\sin A}{\cos A}$

$= \sec A + \tan A$

(viii) $\sin^2 A + \operatorname{cosec}^2 A + 2\sin A \operatorname{cosec} A + \cos^2 A + \sec^2 A + 2\cos A \sec A$

$\Rightarrow 1 + 1 + \cot^2 A + 2\sin A \cdot \dfrac{1}{\sin A} + 1 + \tan^2 A + 2\cos A \cdot \dfrac{1}{\cos A}$

$= 7 + \cot^2 A + \tan^2 A$

(iv) $\dfrac{1+\sec A}{\sec A} = \dfrac{\sin^2 A}{1-\cos A}$ [**Hint :** Simplify LHS and RHS separately]

(v) $\dfrac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A$, using the identity $\operatorname{cosec}^2 A = 1 + \cot^2 A$.

(vi) $\sqrt{\dfrac{1+\sin A}{1-\sin A}} = \sec A + \tan A$

(vii) $\dfrac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$

(viii) $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$

$\frac{1-2\sin\theta}{2\cos\theta-1}$

$\sin^2\theta + \cos^2\theta = 1$

$\cos^2\theta = 1 - \sin^2\theta$

(vii) $\sin\theta(1-2\sin^2\theta) \Rightarrow \sin\theta(1-2\sin^2\theta)$

LHS $= \dfrac{\sin\theta(1-2\sin^2\theta)}{\cos\theta(2\cos^2\theta-1)} = \dfrac{\sin\theta(1-2\sin^2\theta)}{\cos\theta(2-2\sin^2\theta-1)}$

$\Rightarrow \dfrac{\sin\theta(1-2\sin^2\theta)}{\cos\theta[2(1-\sin^2\theta)-1]} = \dfrac{\sin\theta(1-2\sin^2\theta)}{\cos\theta(1-2\sin^2\theta)}$

$= \boxed{\tan\theta} = RHS$

$= \sec\theta + 1$

(ix) $(\text{cosec } A - \sin A)(\sec A - \cos A) = \dfrac{1}{\tan A + \cot A}$

[**Hint :** Simplify LHS and RHS separately]

(x) $\left(\dfrac{1 + \tan^2 A}{1 + \cot^2 A}\right) = \left(\dfrac{1 - \tan A}{1 - \cot A}\right)^2 = \tan^2 A$

(X) $\rightarrow \dfrac{\sec^2 A}{\text{cosec}^2 A}$

$\Rightarrow \dfrac{\frac{1}{\cos^2 A}}{\frac{1}{\sin^2 A}}$

$= \dfrac{\sin^2 A}{\cos^2 A}$

$= \tan^2 A$

$\left(1 - \dfrac{\frac{\sin A}{\cos A}}{1 - \frac{\cos A}{\sin A}}\right)^2$

$\Rightarrow \left(\dfrac{\frac{\cos A - \sin A}{\cos A}}{\frac{\sin A - \cos A}{\sin A}}\right)^2$

$\dfrac{\sin A \cos A - \sin^2 A}{\sin A \cos A - \cos^2 A}$

$= \left[-\dfrac{(\sin A - \cos A)}{\cos A}\right]^2 = \left(\dfrac{-\sin A}{\cos A}\right)^2$

$= \dfrac{\sin^2 A}{\cos^2 A} = \tan^2 A$

$\dfrac{1 \cdot 2 - 3}{1 \cdot 2 - 4} =$

8.5 Summary

In this chapter, you have studied the following points :

1. In a right triangle ABC, right-angled at B,

$$\sin A = \frac{\text{side opposite to angle A}}{\text{hypotenuse}}, \cos A = \frac{\text{side adjacent to angle A}}{\text{hypotenuse}}$$

$$\tan A = \frac{\text{side opposite to angle A}}{\text{side adjacent to angle A}}.$$

2. $\text{cosec } A = \frac{1}{\sin A}$; $\sec A = \frac{1}{\cos A}$; $\tan A = \frac{1}{\cot A}$, $\tan A = \frac{\sin A}{\cos A}$.

3. If one of the trigonometric ratios of an acute angle is known, the remaining trigonometric ratios of the angle can be easily determined.

4. The values of trigonometric ratios for angles $0°$, $30°$, $45°$, $60°$ and $90°$.

5. The value of sin A or cos A never exceeds 1, whereas the value of sec A ($0° \le A < 90°$) or cosec A ($0° < A \le 90°$) is always greater than or equal to 1.

6. $\sin^2 A + \cos^2 A = 1$,

   $\sec^2 A - \tan^2 A = 1$ for $0° \le A < 90°$,

   $\text{cosec}^2 A = 1 + \cot^2 A$ for $0° < A \le 90°$.

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MCQ WORKSHEET-III
CLASS X: CHAPTER - 8
INTRODUCTION TO TRIGONOMETRY

1. In a triangle PQR, right-angled at Q, PR + QR = 25 cm and PQ = 5 cm, then the value of sin P is

(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of the these

2. In a triangle PQR, right-angled at Q, PQ = 3 cm and PR = 6 cm, then $\angle QPR =$

(a) 0° (b) 30° (c) 45° (d) 60°

3. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, then the value of A and B, respectively are

(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these

4. If sin (A – B) = 1 and cos(A + B) = 1, then the value of A and B, respectively are

(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these

5. If $\tan(A - B) = \frac{1}{\sqrt{3}}$ and $\tan(A + B) = \sqrt{3}$, then the value of A and B, respectively are

(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these

10. If $\sin A = \frac{1}{2}$, then the value of $3\cos A - 4\cos^3 A$ is

(a) 0 (b) 1 (c) ½ (d) n.d.

11. If $3\cot A = 4$, then the value of $\cos^2 A - \sin^2 A$ is

(a) $\frac{3}{4}$ (b) $\frac{7}{25}$ (c) ½ (d) $\frac{24}{25}$

12. If $3\tan A = 4$, then the value of $\frac{3\sin A + 2\cos A}{3\sin A - 2\cos A}$ is

(a) 1 (b) $\frac{7}{25}$ (c) 3 (d) $\frac{24}{25}$

$\frac{a}{b}+\frac{c}{d}=\frac{ad+cb}{bd}$

PRACTICE QUESTIONS
CLASS X: CHAPTER - 8
INTRODUCTION TO TRIGONOMETRY
TRIGONOMETRIC IDENTITIES

$\frac{a}{\frac{b}{c}}=\frac{ac}{b}$

1. Prove that $\frac{\cos\theta}{1-\tan\theta}+\frac{\sin\theta}{1-\cot\theta}=\sin\theta+\cos\theta$.

2. Prove that $\frac{1}{2}\left\{\frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}\right\}=\frac{1}{\sin\theta}$.

3. Prove that: $\frac{\tan^3\alpha}{1+\tan^2\alpha}+\frac{\cot^3\alpha}{1+\cot^2\alpha}=\sec\alpha\,\mathrm{cosec}\alpha-2\sin\alpha\cos\alpha$

4. Prove that: $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}=1+\tan A+\cot A=1+\sec A\,\mathrm{cosec}A$.

5. Prove that:. $\frac{1+\sin A-\cos A}{1+\sin A+\cos A}=\sqrt{\frac{1-\cos A}{1+\cos A}}$

6. Prove that $(\tan A+\mathrm{cosec}B)^2-(\cot B-\sec A)^2=2\tan A\cot B(\mathrm{cosec}A+\sec B)$.

7. Prove that: $\frac{\cos A}{1+\sin A}+\frac{1+\sin A}{\cos A}=2\sec A$.

LHS =

(1) $\frac{\cos\theta}{1-\frac{\sin\theta}{\cos\theta}}+\frac{\sin\theta}{1-\frac{\cos\theta}{\sin\theta}}$

$=\frac{\cos\theta}{\left(\frac{\cos\theta-\sin\theta}{\cos\theta}\right)}+\frac{\sin\theta}{\left(\frac{\sin\theta-\cos\theta}{\sin\theta}\right)}$

$=\frac{\cos^2\theta}{\cos\theta-\sin\theta}+\frac{\sin^2\theta}{\sin\theta-\cos\theta}$

$=\frac{\cos^2\theta}{\cos\theta-\sin\theta}-\frac{\sin^2\theta}{\cos\theta-\sin\theta}$

$=\frac{1}{\cos\theta-\sin\theta}\left[\cos^2\theta-\sin^2\theta\right]$

$=\frac{1}{\cos\theta-\sin\theta}\left[(\cos\theta+\sin\theta)(\cos\theta-\sin\theta)\right]=\cos\theta+\sin\theta$

$=$ RHS

$\frac{a}{b}+\frac{c}{d}=\frac{ad+cb}{bd}$

PRACTICE QUESTIONS
CLASS X: CHAPTER - 8
INTRODUCTION TO TRIGONOMETRY
TRIGONOMETRIC IDENTITIES

1. Prove that $\frac{\cos\theta}{1-\tan\theta}+\frac{\sin\theta}{1-\cot\theta}=\sin\theta+\cos\theta$.

2. Prove that $\frac{1}{2}\left\{\frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}\right\}=\frac{1}{\sin\theta}$.

3. Prove that: $\frac{\tan^3\alpha}{1+\tan^2\alpha}+\frac{\cot^3\alpha}{1+\cot^2\alpha}=\sec\alpha\,\mathrm{cosec}\alpha-2\sin\alpha\cos\alpha$

4. Prove that: $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}=1+\tan A+\cot A=1+\sec A\,\mathrm{cosec}A$.

5. Prove that:. $\frac{1+\sin A-\cos A}{1+\sin A+\cos A}=\sqrt{\frac{1-\cos A}{1+\cos A}}$

6. Prove that $(\tan A+\mathrm{cosec}B)^2-(\cot B-\sec A)^2=2\tan A\cot B(\mathrm{cosec}A+\sec B)$.

7. Prove that: $\frac{\cos A}{1+\sin A}+\frac{1+\sin A}{\cos A}=2\sec A$.

② $\frac{1}{2}\left[\frac{\sin^2\theta+(1+\cos\theta)^2}{(1+\cos\theta)\sin\theta}\right]$

$=\frac{1}{2}\left[\frac{\sin^2\theta+1+\cos^2\theta+2\cos\theta}{(1+\cos\theta)(\sin\theta)}\right]$

$=\frac{1}{2}\left[\frac{1+1+2\cos\theta}{(1+\cos\theta)\sin\theta}\right]$

$=\frac{1}{2}\left[\frac{2+2\cos\theta}{(1+\cos\theta)\sin\theta}\right]$

$=\frac{1}{2}\times\left[\frac{2\times(1+\cos\theta)}{(1+\cos\theta)(\sin\theta)}\right]$

$=\frac{1}{\sin\theta}=\mathrm{RHS}$

PRACTICE QUESTIONS
CLASS X: CHAPTER - 8
INTRODUCTION TO TRIGONOMETRY
TRIGONOMETRIC IDENTITIES

1. Prove that $\frac{\cos\theta}{1-\tan\theta}+\frac{\sin\theta}{1-\cot\theta}=\sin\theta+\cos\theta$.

2. Prove that $\frac{1}{2}\left\{\frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}\right\}=\frac{1}{\sin\theta}$.

3. Prove that: $\frac{\tan^3\alpha}{1+\tan^2\alpha}+\frac{\cot^3\alpha}{1+\cot^2\alpha}=\sec\alpha\,\mathrm{cosec}\alpha-2\sin\alpha\cos\alpha$

4. Prove that: $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}=1+\tan A+\cot A=1+\sec A\,\mathrm{cosec}A$.

5. Prove that: $\frac{(1+\sin A)-\cos A}{(1+\sin A)+\cos A}=\sqrt{\frac{1-\cos A}{1+\cos A}}$

6. Prove that $(\tan A+\mathrm{cosec}B)^2-(\cot B-\sec A)^2=2\tan A\cot B(\mathrm{cosec}A+\sec B)$.

7. Prove that: $\frac{\cos A}{1+\sin A}+\frac{1+\sin A}{\cos A}=2\sec A$.

⑤ LHS $=\frac{(1+\sin A)-\cos A}{(1+\sin A)+\cos A}\times\frac{(1+\sin A)-\cos A}{(1+\sin A)-\cos A}$

$=\frac{[(1+\sin A)-\cos A]^2}{(1+\sin A)^2-\cos^2 A}$

$=\frac{(1+\sin A)^2+\cos^2 A-2(1+\sin A)\cos A}{1+\sin^2 A+2\sin A-\cos^2 A}$

$=\frac{1+\sin^2 A+2\sin A+\cos^2 A-2\cos A-2\cos A\sin A}{1+\sin^2 A+2\sin A-\cos^2 A}$

$=\frac{2+2\sin A-2\cos A-2\cos A\sin A}{\sin^2 A+\sin^2 A+2\sin A}$

$=\frac{2(1+\sin A-\cos A-\cos A\sin A)}{2\sin A(\sin A+1)}$

$\Rightarrow\frac{(1+\sin A)-\cos A(1+\sin A)}{\sin A(\sin A+1)}$

$\Rightarrow\frac{(1+\sin A)(1-\cos A)}{\sin A(1+\sin A)}$

$=\frac{1-\cos A}{\sin A}$

RHS $=\sqrt{\frac{1-\cos A}{1+\cos A}\times\frac{1-\cos A}{1-\cos A}}$

$=\sqrt{\frac{(1-\cos A)^2}{1-\cos^2 A}}=\sqrt{\frac{(1-\cos A)^2}{\sin^2 A}}=\frac{1-\cos A}{\sin A}$

8. Prove that: $\dfrac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A$.

9. Prove that: $\dfrac{\sin A + \cos A}{\sin A - \cos A} + \dfrac{\sin A - \cos A}{\sin A + \cos A} = \dfrac{2}{\sin^2 A - \cos^2 A} = \dfrac{2}{2\sin^2 A - 1} = \dfrac{2}{1 - 2\cos^2 A}$.

10. Prove that $\dfrac{\sin A}{\cot A + \operatorname{cosec} A} = 2 + \dfrac{\sin A}{\cot A - \operatorname{cosec} A}$.

11. Prove that $\sqrt{\dfrac{\sec A - 1}{\sec A + 1}} + \sqrt{\dfrac{\sec A + 1}{\sec A - 1}} = 2\operatorname{cosec} A$.

12. Prove that: $\dfrac{1}{\operatorname{cosec} A - \cot A} - \dfrac{1}{\sin A} = \dfrac{1}{\sin A} - \dfrac{1}{\operatorname{cosec} A + \cot A}$.

13. Prove that: $\dfrac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} = \sec\theta + \tan\theta = \dfrac{1 + \sin\theta}{\cos\theta}$

14. If $x = a\sin\theta + b\cos\theta$ and $y = a\cos\theta + b\sin\theta$, prove that $x^2 + y^2 = a^2 + b^2$.

15. If $\sec\theta + \tan\theta = m$, show that $\left(\dfrac{m^2 - 1}{m^2 + 1}\right) = \sin\theta$.